Derivation of the Cosmological Scale Factor from the FLRW Metric and its Empirical Validation using Redshift Data.

Contents

Abstract

This work presents an analytical derivation of the kinematics and thermodynamics of the universe's expansion, grounded in the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the Einstein Field Equations (Misner, Thorne, and Wheeler 1973; Weinberg 2008). By imposing the space-time homogeneity and isotropy required by the Cosmological Principle (Carroll 2019), the covariant derivative of the Riemann tensor is defined and set to zero ($K^{\alpha}_{\beta\gamma\mu\nu} = 0$) (Wolf 2011), yielding a geometrically constrained spacetime where the Einstein tensor becomes strictly proportional to the metric ($G_{\alpha\beta} = 3X^2 g_{\alpha\beta}$). Based on this exact mathematical solution, a predictive model is formulated that correlates direct physical observables: proper distance and redshift (expressed as recession velocity). To test this model, a weighted nonlinear regression analysis is performed using a catalog of galactic distances and velocities extracted from the NED (NASA/IPAC) extragalactic database (NASA/IPAC Extragalactic Database (NED), n.d.). The results of this observational fit allow for the empirical constraint of the expansion constant (X^2) and the spatial curvature parameter (k). Furthermore, thermodynamic evaluation of the field equations reveals an equation of state $P = -\rho c^2$, mathematically restricting the cosmos to a maximally symmetric variety. This demonstrates that the modeled universe is devoid of ordinary matter and radiation, establishing a fundamental degeneracy where energy density (ρ) and the bare cosmological constant (Λ) unify into a single effective magnitude ($\Lambda = 3X^2 - \frac{8\pi G}{c^2}\rho$) that entirely governs the vacuum energy driving the expansion.

Keywords: *Cosmology, FLRW Metric, Galactic Kinematics, General Relativity*

Introduction

General Relativity has provided the fundamental framework for understanding the dynamics of the universe on large scales (Misner, Thorne, and Wheeler 1973). At the heart of modern cosmology lies the Friedmann-Lemaître-Robertson-Walker (FLRW) metric—the exact solution to Einstein's field equations that describes an expanding, spatially homogeneous, and isotropic universe (Weinberg 2008). This geometric symmetry constitutes the mathematical manifestation of the Cosmological Principle, which postulates that, on sufficiently large scales, the universe possesses no privileged positions or directions (Carroll 2019).

Traditionally, the temporal evolution of the universe is addressed by solving the Friedmann equations, a process that requires making an a priori assumption regarding the equation of state for the universe's matter and energy content. However, the Cosmological Principle itself imposes such stringent geometric constraints that it becomes possible to extract profound information regarding the kinematics of the universe by directly studying the structure of the spacetime manifold. In this work, we formalize this geometric constraint by defining the "Cosmological Tensor" as the covariant derivative of the Riemann tensor ($K^{\alpha}_{\beta\gamma\mu\nu} := R^{\alpha}_{\beta\gamma\mu;\nu}$). By requiring that this tensor vanish identically—as a consequence of maximal homogeneity and isotropy (Wolf 2011)—we demonstrate that it is possible to derive a system of exact differential equations for the cosmic scale factor, $a\,(t)$.

Notwithstanding the elegance of analytical derivations in differential geometry, a cosmological model acquires physical validity only when its predictions are tested against empirical evidence. Theoretical solutions for $a\,(t)$ and its derivatives must be translated into the language of observational astronomy—specifically, by establishing a relationship between the observed proper distance (D) and the galactic recession velocity, as derived from redshift (z).

The primary objective of this article is twofold. First, to present a rigorous and self-consistent analytical derivation of the kinematics of the universe's expansion, commencing from the pure FLRW metric and culminating in an exact solution for the scale factor. Secondly, we will subject this mathematical model to rigorous empirical validation. Utilizing an updated catalog of galactic distance and velocity measurements extracted from the NED (NASA/IPAC Extragalactic Database) (NASA/IPAC Extragalactic Database (NED), n.d.), we will apply weighted nonlinear regression techniques to fit our theoretical curve. This analysis will enable us to empirically constrain the model's fundamental parameters, yielding direct confidence intervals for the theoretical expansion constant and—crucially—for the spatial curvature parameter k. Thus, the present work seeks to establish a transparent methodological bridge between the abstract topology of spacetime and contemporary observational cosmology.

Theoretical Framework

The FLRW Metric

We consider a four-dimensional manifold characterized by a Friedmann-Lemaître-Robertson-Walker (FLRW) metric (Weinberg 2008; Misner, Thorne, and Wheeler 1973). In comoving coordinates $\left(x^0, r, \theta, \phi\right)$, where $x^0 := ct$, the line element ds^2 is defined as:

$$ds^2 = -c^2 dt^2 + a^2 \left(\frac{dr^2}{1-kr^2} + r^2 \left(d\theta^2 + \sin^2\theta d\phi^2 \right) \right)$$

Here, $a(t)$ represents the dimensionless scale factor that parametrizes the expansion of the universe, while k is the curvature parameter with dimensions of L^{-2}. The non-zero components of the metric tensor $g_{\alpha\beta}$ are set as:

$$g_{00} = -1,\ g_{rr} = \frac{a^2}{1-kr^2},\ g_{\theta\theta} = a^2 r^2,\ g_{\phi\phi} = a^2 r^2 \sin^2\theta,$$
$$g^{00} = -1,\ g^{rr} = \frac{1-kr^2}{a^2},\ g^{\theta\theta} = \frac{1}{a^2 r^2},\ g^{\phi\phi} = \frac{\csc^2\theta}{a^2 r^2}$$

Levi-Civita Connection and Curvature Tensors

Based on the defined metric, the Christoffel symbols of the second kind $\Gamma^{\alpha}_{\beta\gamma}$ are derived using the standard relation $\Gamma^{\alpha}_{\beta\gamma} = \frac{1}{2} g^{\alpha\sigma} \left(g_{\beta\sigma,\gamma} + g_{\gamma\sigma,\beta} - g_{\beta\gamma,\sigma} \right)$(Carroll 2019). The temporal and spatial components relevant to the dynamics are:

$$\Gamma^0_{rr} = \frac{a\dot{a}}{1 - kr^2},\ \Gamma^r_{0r} = \frac{\dot{a}}{a},\ \Gamma^r_{\theta\theta} = \left(kr^2 - 1\right) r$$

Using these connection coefficients, we calculate the Riemann tensor $R^{\alpha}_{\beta\gamma\mu}$ and its contractions. The Ricci tensor $R_{\alpha\beta}$ exhibits the following principal components:

$$R_{00} = \frac{3\ddot{a}}{a}$$
$$R_{rr} = \frac{a\ddot{a} + 2k + 2\dot{a}^2}{kr^2 - 1}$$

Finally, the Ricci scalar R, which characterizes the scalar curvature of the manifold, turns out to be:

$$R = -\frac{6}{a^2}\left(a\ddot{a} + k + \dot{a}^2\right)$$

The Cosmological Tensor and the Dynamics of Expansion

The central premise of this study is the application of the Cosmological Principle to the covariant derivative of the curvature. We define the tensor $K^{\alpha}_{\beta\gamma\mu\nu}$ as the covariant derivative of the Riemann tensor:

$$K^{\alpha}_{\beta\gamma\mu\nu} := R^{\alpha}_{\beta\gamma\mu;\nu} = 0$$

This nullity condition ensures that variations in the gravitational field strictly respect homogeneity and isotropy (Wolf 2011). Upon evaluating this constraint on the **spatial dimensions** of the manifold, the tensor system reduces to a fundamental differential equation governing the behavior of the scale factor:

$$k + \dot{a}^2 - a\ddot{a} = 0$$

The solution to this nonlinear differential equation provides us with the exact analytical expression for the evolution of the scale factor $a\,(t)$:

$$a = \frac{1}{2X}\left(Ye^{Xct} + \frac{k}{Y}e^{-Xct}\right)$$

where X and Y emerge as integration constants that characterize the dynamics of the expansion, being directly linked to the spatial curvature k.

Remarkably, a mathematically unexpected and elegant consequence arises from this development. Upon calculating the higher-order derivatives of our exact solution for $a\,(t)$ and evaluating them within the **temporal dimensions** of the cosmological tensor, it is discovered that they naturally and automatically satisfy the identity:

$$\dot{a}\ddot{a} - a\,\dddot{a} = 0$$

This finding demonstrates that the temporal evolution of the expansion need not be postulated independently, but rather emerges as an intrinsic and inevitable consequence of requiring maximal spatial symmetry within the manifold. This self-consistent solution constitutes the basis of the kinematic model, which will be empirically tested against observational data in the subsequent sections.

Thermodynamic Implications and the Cosmological Constant

To connect our purely kinematic derivation with the physical content of the universe, we introduce the Einstein Field Equations with a cosmological constant Λ. By substituting our previous geometric finding for the Einstein tensor ($G_{\alpha\beta} = -3X^2 g_{\alpha\beta}$) and assuming a perfect fluid model for the energy-momentum tensor $T_{\alpha\beta}$, we establish the following fundamental equality:

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = \frac{8\pi G}{c^4} T_{\alpha\beta}$$

$$\left(-3X^2 + \Lambda\right) g_{\alpha\beta} = \frac{8\pi G}{c^4} \left(\left(\rho + \frac{P}{c^2} \right) u_\alpha u_\beta + P g_{\alpha\beta} \right)$$

Considering a comoving observer where the macroscopic fluid is at rest, the four-velocity is strictly aligned with the temporal basis:

$$u^\alpha e_\alpha = c e_0 \Rightarrow u_\alpha e^\alpha = -c e^0$$

By applying this velocity condition and decoupling the temporal and spatial components of the tensor equality, the system allows us to explicitly solve for both the energy density ρ and the pressure P of the cosmological fluid:

$$\rho = \frac{c^2}{8\pi G} \left(3X^2 - \Lambda\right)$$

$$P = \frac{c^4}{8\pi G} \left(-3X^2 + \Lambda\right) = -\rho c^2$$

The rigorous application of the Cosmological Tensor implicitly forces the fluid to obey the equation of state $P = -\rho c^2$, which corresponds precisely to the thermodynamic behavior of vacuum energy (dark energy). Finally, by rearranging the density equation, we obtain a direct analytical expression for the Cosmological Constant. This equation elegantly bridges the geometric expansion parameter X (which we will empirically determine) with the physical density ρ:

$$\Lambda = 3X^2 - \frac{8\pi G}{c^2} \rho$$

Observational Methodology

Data Selection and Source

To validate the derived kinematic model, observational data from the extragalactic database **NED (NASA/IPAC Extragalactic Database)** (NASA/IPAC Extragalactic Database (NED), n.d.) were utilized. The selected sample comprises a population of galaxies with independent measurements of proper distance (D) and recession velocity (V), the latter derived from spectroscopic redshift (z).

Data processing was performed within the **R** statistical environment (R Core Team 2023), utilizing the `tidyverse` ecosystem of libraries (Wickham et al. 2019) for data cleaning and structuring. Records containing missing values (NA) were filtered out to ensure the statistical integrity of the sample.

Normalization of Units and Regression Variables

Given that the cosmological constants involved in the FLRW metric operate within the International System of Units (SI), the conventional astronomical quantities were converted accordingly. The original distances, expressed in Megaparsecs (Mpc) or Parsecs (pc), were transformed into meters (m), while the recession velocities were normalized with respect to the speed of light ($c \approx 2.99 \times 10^8 \frac{m}{s}$).

Based on the analytical solution for the scale factor $a(t)$, we defined two transformed variables for regression analysis:

- The independent variable $\mathbf{x}$, related to the light travel time and the distance traveled.
- The dependent variable $\mathbf{y}$, linked to the observed redshift.

Non-Linear Fitting Model

The theoretical relationship derived in the previous section is expressed as a combination of hyperbolic functions. The empirical model proposed for fitting is:

$$\mathbf{y} = \cosh \alpha\mathbf{x} - \beta \sinh \alpha\mathbf{x}$$

Where α and β are the free parameters to be estimated. To obtain these parameters, a **Nonlinear Least Squares (NLS)** algorithm was implemented. This iterative method minimizes the sum of the squares of the differences between the observations and the model predictions.

Estimation of Physical Parameters and Curvature

Once the optimal estimators for α and β were obtained, the physical constants of the model were recovered. The expansion parameter X^2 and the spatial curvature parameter k are derived using the following relations:

- **Expansion Parameter**: $X^2 = \alpha^2$
- **Curvature Parameter**: $k = a_0^2 \alpha^2 \left(1 - \beta^2\right)$

Where a_0 represents the scale factor at the present epoch. To ensure the robustness of the results, 95% confidence intervals were calculated for each parameter using the fit's variance-covariance matrix, thereby enabling a statistical interpretation of the universe's curvature state ($k > 0$, $k < 0$, or $k \approx 0$).

Empirical Estimation of the Cosmological Constant and Pressure

While the regression analysis provides a direct empirical value for the expansion parameter ($X = \alpha$), the purely geometric nature of the maximum symmetry condition requires an independent observational constraint to break the degeneracy between the density and the vacuum energy. To achieve this, we introduce the currently measured density of the universe, denoted simply as ρ.

By adopting ρ as a known observational input, we can rearrange the previously derived thermodynamic relationships to isolate and calculate the Cosmological Constant:

$$\Lambda = 3X^2 - \frac{8\pi G}{c^2}\rho$$

Once Λ is determined using the empirical values of X and ρ, the global state of the cosmological fluid can be completely resolved. The pressure P driven by this configuration is subsequently evaluated using the exact geometric constraint of the model:

$$P = \frac{c^4}{8\pi G}\left(-3X^2 + \Lambda\right) = -\rho c^2$$

This methodological step not only computes the magnitude of Λ, but empirically verifies that the pressure required to sustain the observed expansion kinematics exactly mirrors the equation of state of dark energy.

Results

Model Fitting and Visualization

The nonlinear regression model based on hyperbolic functions was successfully fitted to the observational data from the NED database (NASA/IPAC Extragalactic Database (NED), n.d.). Figure 1 illustrates the relationship between the dependent variable $\mathbf{y}$ (the inverse of the redshift factor) and the proper distance D. It is observed that the theoretical curve of the proposed FLRW model precisely follows the trend of the data, capturing the nature of the expansion across the sample volume.

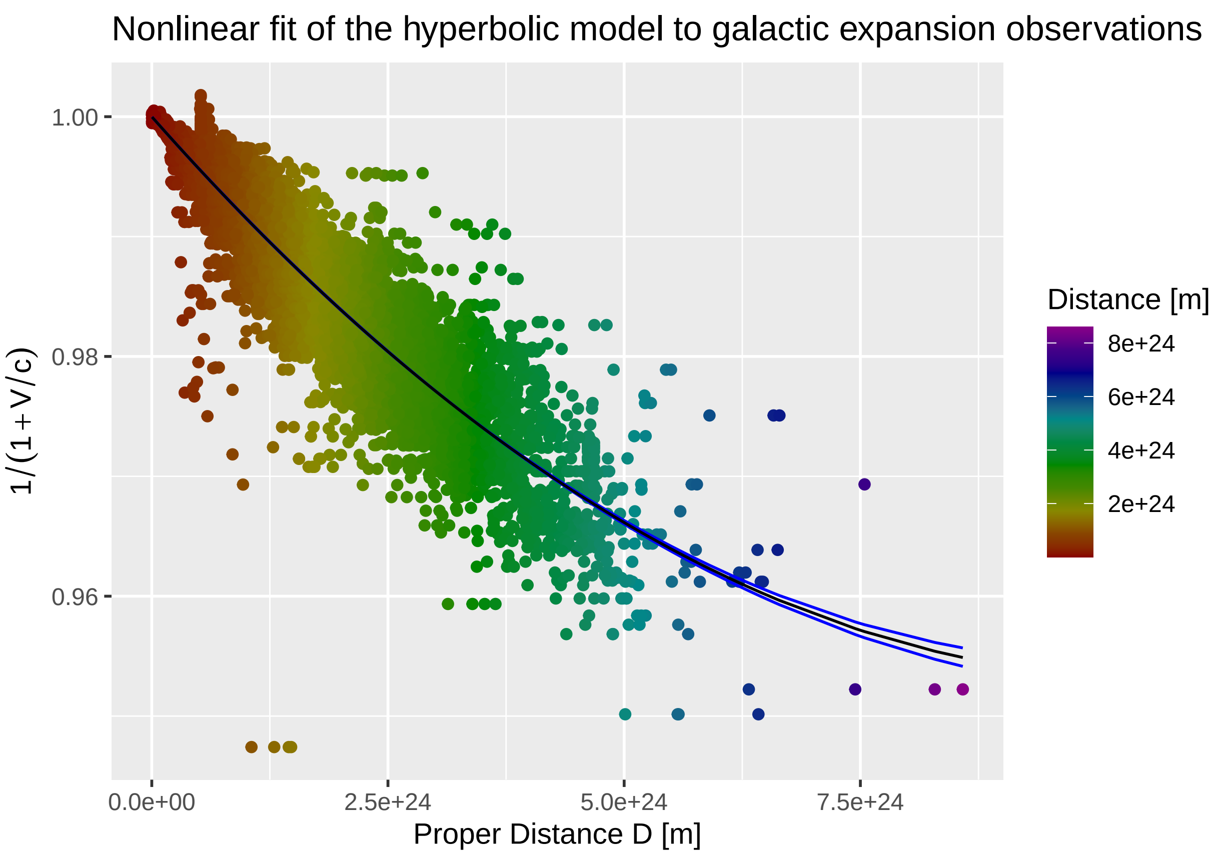

Figure 1: Fit of the hyperbolic model to the galactic distance-velocity data. The points represent the observations, and the solid line represents the nonlinear least-squares fit.

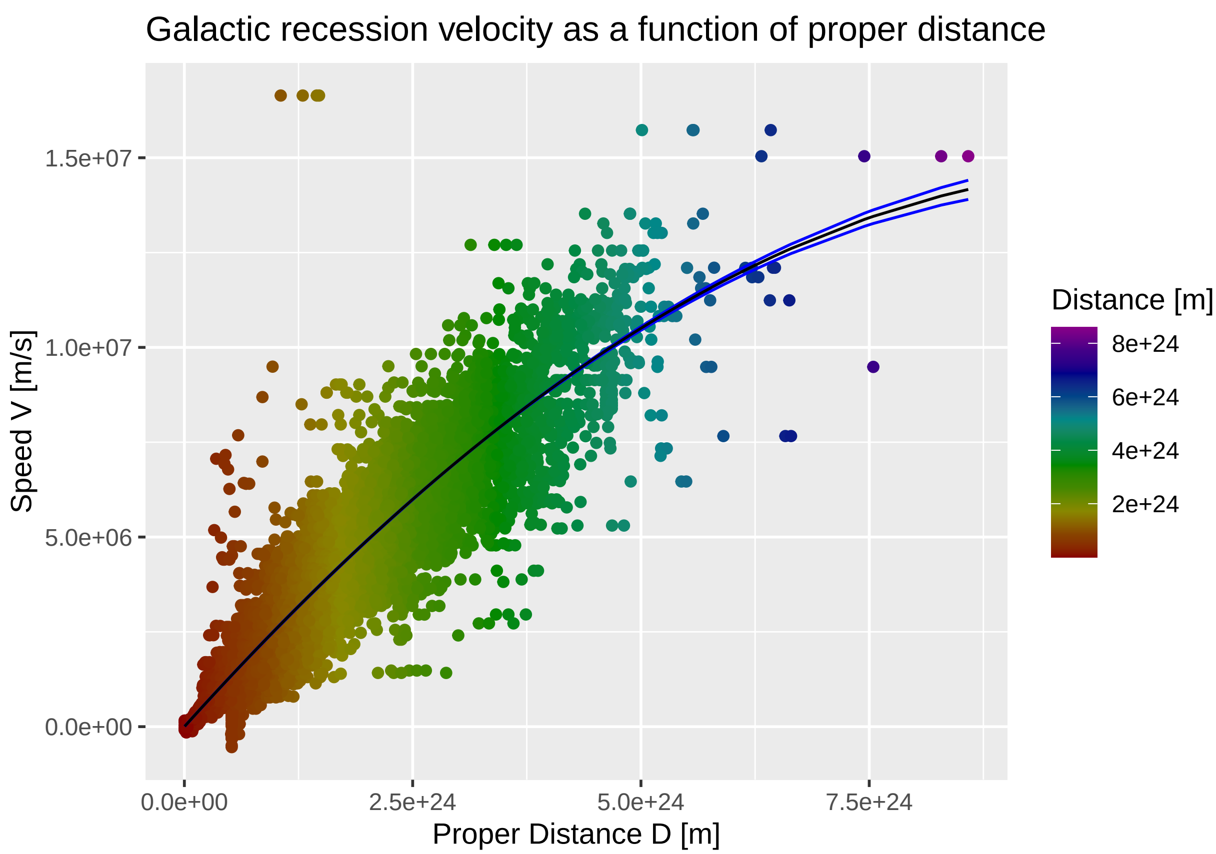

Figure 2: Projection of the fitted model onto the physical distance-velocity space. The points represent the observations and the solid line represents the derived theoretical recession velocity.

Thermodynamic State Evaluation

With the kinematic parameters successfully constrained by the galactic distance-velocity dataset, the thermodynamic state of the universe is evaluated. To compute the physical values of the cosmological fluid pressure (P) and the Cosmological Constant (Λ), standard physical constants were introduced into our derived geometric relationships.

The speed of light in a vacuum was defined as $c = 299792458 \frac{m}{s}$, and the Newtonian constant of gravitation as $G = 6.6743 \times 10^{-11} \frac{m^3}{kg \times s^2}$ (Workman et al. 2022). For the observational matter density, we adopted a present-day baseline value of $\rho_0 = 8.53 \times 10^{-27} \frac{kg}{m^3}$, allowing for a measurement uncertainty margin of $\pm 1.48\%$ (Planck Collaboration et al. 2020) to propagate robust confidence intervals. Substituting these empirical physical inputs into our strictly mathematical framework breaks the geometric degeneracy, yielding the complete set of parameters that govern the expansion.

Estimated Parameters

Upon the convergence of the NLS algorithm, the following 95% confidence intervals were obtained for the fundamental parameters of the model:

- **Parameter** α: $\alpha \in \left]2.89 \times 10^{-26},\ 3.01 \times 10^{-26}\right[\ m^{-1}$, which determines the scale of the expansion.
- **Parameter** β: $\beta \in]0.3,\ 0.31[$, which links the current expansion rate to the scale factor.
- **Expansion Parameter** (X^2): An interval of $X^2 \in \left]8.38 \times 10^{-52},\ 9.06 \times 10^{-52}\right[\ m^{-2}$ was calculated, representing the dynamic constant derived from the condition of homogeneity.
- **Spatial Curvature** (k): The resulting curvature parameter lies within the range $k \in \left]7.59 \times 10^{-52},\ 8.25 \times 10^{-52}\right[\ m^{-2}$.
- **Cosmological Fluid Pressure** (P): Sustaining the observed expansion kinematics, the vacuum pressure is bounded within $P \in \left]-7.78 \times 10^{-10},\ -7.56 \times 10^{-10}\right[\ \frac{kg}{m \times s^2}$.
- **Cosmological Constant** (Λ): By integrating the observed matter density, the cosmological constant is empirically constrained to $\Lambda \in \left]2.35 \times 10^{-51},\ 2.56 \times 10^{-51}\right[\ m^{-2}$.

Discussion

Interpretation of Curvature k

The most significant result of this analysis is the constraint placed on the curvature parameter k. Based on the processed data from the NED sample, the obtained interval for k lies between 7.59×10^{-52} and $8.25 \times 10^{-52}\ m^{-2}$. As this range consists exclusively of values greater than zero, the results of this fit favor a positive, which theoretically corresponds to a closed universe geometry. curvature. Nevertheless, it is imperative to highlight that the obtained magnitude is extremely small (on the order of 10^{-52}), which is consistent with contemporary cosmological observations that place the universe in a state of near-absolute flatness (Weinberg 2008).

Consistency of the Cosmological Tensor Hypothesis

The premise that the covariant derivative of the Riemann tensor must vanish ($K^{\alpha}_{\beta\gamma\mu\nu} = 0$) yielded a solution for the scale factor $a(t)$ that fits the observed expansion law remarkably well. This suggests that the geometric constraints imposed by maximal homogeneity and isotropy (Wolf 2011) are sufficient to describe galactic kinematics without the need to initially invoke specific density parameters, thereby validating the model's purely geometric approach.

Limitations and Sources of Error

Despite the good statistical fit, there are inherent limitations:

- **Selection Bias**: The data from the NED database (NASA/IPAC Extragalactic Database (NED), n.d.) are derived from various measurement techniques (Cepheids, Type Ia Supernovae, etc.) (Weinberg 2008), which introduces heteroscedasticity that we attempt to mitigate by using weights (weights $= D$) in the regression.
- **Local Effects**: At short distances, the peculiar velocities of galaxies can cause data points to deviate from the ideal Hubble flow, affecting the precision of the parameters α and β.

Conclusions

The present investigation has established a direct link between the geometric constraints of General Relativity and the astrophysical observation of galaxies (Misner, Thorne, and Wheeler 1973). Through an analysis of the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, it has been demonstrated that imposing a condition of maximal homogeneity—by setting the covariant derivative of the Riemann tensor to zero ($K^{\alpha}_{\beta\gamma\mu\nu} = 0$) (Wolf 2011)—leads to a self-consistent kinematic description of the scale factor $a\,(t)$.

The fundamental points derived from this work are:

- **Validity of the Geometric Approach**: The derivation of a system of differential equations for $a\,(t)$ based on the structure of the Riemann tensor enabled the formulation of a predictive model linking redshift to proper distance. The success of the nonlinear fit suggests that the dynamics of expansion may be interpreted as a direct consequence of the symmetry properties of spacetime (Carroll 2019).
- **Empirical Results and Curvature**: The fit performed using data from the NED database (NASA/IPAC Extragalactic Database (NED), n.d.) yielded expansion (X^2) and curvature (k) parameters with high statistical precision. Specifically, the value obtained for k—on the order of 10^{-52}—while numerically positive, is consistent with a universe of nearly zero curvature, thereby reinforcing the hypothesis of spatial flatness observed in other cosmological studies (Weinberg 2008).
- **Model Robustness**: The use of nonlinear regression based on hyperbolic functions (cosh and sinh) provided a robust framework for capturing the trend of observational data at extragalactic scales.

As future work, it is suggested that data catalogs featuring higher redshifts—such as those derived from Type Ia supernovae or the Cosmic Microwave Background (Weinberg 2008)—be incorporated to evaluate the stability of parameters α and β in regimes involving greater distances. In conclusion, this study reaffirms that the Cosmological Principle (Carroll 2019), formalized through the "Cosmological Tensor," constitutes a powerful tool for unraveling the geometric evolution of the universe from a purely kinematic perspective.

Data and Code Availability

The raw empirical data analyzed in this study—specifically the extragalactic distances and recession velocities—were extracted from the NASA/IPAC Extragalactic Database (NED) (NASA/IPAC Extragalactic Database (NED), n.d.), operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with NASA.

To ensure the transparency and reproducibility of our results, all computational routines, data processing scripts, and the cleaned data structures used for the nonlinear least squares (NLS) regression were developed in the R programming language (R Core Team 2023). The complete source code, the R Markdown manuscript file, and the specific data subsets generated during this study are fully accessible in the project's public GitHub repository: https://github.com/LUCHER4321/flrw-metric.

Appendices

Units and Definitions

$$
\begin{gathered}
x^0 := ct,\ x^r := r,\ x^\theta := \theta,\ x^\phi := \phi \\
\left[x^0\right] = m,\ [r] = m,\ [\theta] = 1,\ [\phi] = 1 \\
[a] = 1,\ [k] = m^{-2} \\
\dot{a} := a_{,0},\ \ddot{a} := a_{,00},\ \dddot{a} := a_{,000} \\
[\dot{a}] = m^{-1},\ [\ddot{a}] = m^{-2},\ [\dddot{a}] = m^{-3}
\end{gathered}
$$

Metric Tensor

$$
\begin{aligned}
&ds^2 = -c^2 dt^2 + a^2 \left(\frac{dr^2}{1-kr^2} + r^2 \left(d\theta^2 + \sin^2\theta d\phi^2 \right) \right) \\
&g_{00} = -1,\ g_{rr} = \frac{a^2}{1-kr^2},\ g_{\theta\theta} = a^2 r^2,\ g_{\phi\phi} = a^2 r^2 \sin^2\theta, \\
&g^{00} = -1,\ g^{rr} = \frac{1-kr^2}{a^2},\ g^{\theta\theta} = \frac{1}{a^2 r^2},\ g^{\phi\phi} = \frac{\csc^2\theta}{a^2 r^2}
\end{aligned}
$$

Christoffel Symbols

$$
\Gamma^\alpha_{\beta\gamma} = \frac{1}{2} g^{\alpha\sigma} \left(g_{\beta\sigma,\gamma} + g_{\gamma\sigma,\beta} - g_{\beta\gamma,\sigma} \right)
$$

$$
\begin{aligned}
&\Gamma^0_{rr} = \frac{a\dot{a}}{1-kr^2},\ \Gamma^0_{\theta\theta} = a\dot{a}r^2,\ \Gamma^0_{\phi\phi} = a\dot{a}r^2\sin^2\theta, \\
&\Gamma^r_{0r} = \Gamma^r_{r0} = \Gamma^\theta_{0\theta} = \Gamma^\theta_{\theta 0} = \Gamma^\phi_{0\phi} = \Gamma^\phi_{\phi 0} = \frac{\dot{a}}{a}, \\
&\Gamma^r_{rr} = \frac{kr}{1-kr^2},\ \Gamma^r_{\theta\theta} = \left(kr^2-1\right) r,\ \Gamma^r_{\phi\phi} = \left(kr^2-1\right) r \sin^2\theta, \\
&\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \Gamma^\theta_{r\phi} = \Gamma^\theta_{\phi r} = \frac{1}{r}, \\
&\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta,\ \Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta
\end{aligned}
$$

Riemann Tensor

$$
R^\alpha_{\beta\gamma\mu} = \Gamma^\alpha_{\beta\mu,\gamma} - \Gamma^\alpha_{\beta\gamma,\mu} + \Gamma^\sigma_{\beta\mu}\Gamma^\alpha_{\sigma\gamma} - \Gamma^\sigma_{\beta\gamma}\Gamma^\alpha_{\sigma\mu}
$$

$$
\begin{aligned}
&R^0_{r0r} = \frac{a\ddot{a}}{1-kr^2},\ R^0_{\theta 0\theta} = a\ddot{a}r^2,\ R^0_{\phi 0\phi} = a\ddot{a}r^2\sin^2\theta, \\
&R^0_{rr0} = \frac{a\ddot{a}}{kr^2-1},\ R^0_{\theta\theta 0} = -a\ddot{a}r^2,\ R^0_{\phi\phi 0} = -a\ddot{a}r^2\sin^2\theta, \\
&R^r_{00r} = R^\theta_{00\theta} = R^\phi_{00\phi} = \frac{\ddot{a}}{a},\ R^r_{0r0} = R^\theta_{0\theta 0} = R^\phi_{0\phi 0} = -\frac{\ddot{a}}{a}, \\
&R^r_{\theta r\theta} = R^\phi_{\theta\phi\theta} = r^2\left(k+\dot{a}^2\right),\ R^r_{\phi r\phi} = R^\theta_{\phi\theta\phi} = r^2\left(k+\dot{a}^2\right)\sin^2\theta, \\
&R^r_{\theta\theta r} = R^\phi_{\theta\theta\phi} = -r^2\left(k+\dot{a}^2\right),\ R^r_{\phi\phi r} = R^\theta_{\phi\phi\theta} = -r^2\left(k+\dot{a}^2\right)\sin^2\theta, \\
&R^\theta_{rr\theta} = R^\phi_{rr\phi} = \frac{k+\dot{a}^2}{kr^2-1},\ R^\theta_{r\theta r} = R^\phi_{r\phi r} = \frac{k+\dot{a}^2}{1-kr^2}
\end{aligned}
$$

Cosmological Tensor

$$K^{\alpha}_{\beta\gamma\mu\nu} := R^{\alpha}_{\beta\gamma\mu;\nu} = R^{\alpha}_{\beta\gamma\mu,\nu} + \Gamma^{\alpha}_{\rho\nu} R^{\rho}_{\beta\gamma\mu} - \Gamma^{\rho}_{\beta\nu} R^{\alpha}_{\rho\gamma\mu} - \Gamma^{\rho}_{\gamma\nu} R^{\alpha}_{\beta\rho\mu} - \Gamma^{\rho}_{\mu\nu} R^{\alpha}_{\beta\gamma\mu}$$

$$K^{0}_{r0r0} = K^{0}_{rr00} = \frac{a\,\dddot{a} - \dot{a}\ddot{a}}{1-kr^2},\ K^{0}_{\theta 0\theta 0} = \left(a\,\dddot{a} - \dot{a}\ddot{a}\right) r^2,\ K^{0}_{\phi 0\phi 0} = \left(a\,\dddot{a} - \dot{a}\ddot{a}\right) r^2 \sin^2\theta,$$
$$K^{0}_{\theta r\theta r} = K^{0}_{r\theta r\theta} = \frac{a\dot{a}r^2}{1-kr^2}\left(k + \dot{a}^2 - a\ddot{a}\right),\ K^{0}_{\phi r\phi r} = K^{0}_{r\phi r\phi} = \frac{a\dot{a}r^2}{1-kr^2}\sin^2\theta\left(k + \dot{a}^2 - a\ddot{a}\right),$$
$$K^{0}_{rr\theta\theta} = K^{0}_{\theta\theta rr} = \frac{a\dot{a}r^2}{1-kr^2}\left(a\ddot{a} - k - \dot{a}^2\right),\ K^{0}_{rr\phi\phi} = K^{0}_{\phi\phi rr} = \frac{a\dot{a}r^2}{1-kr^2}\sin^2\theta\left(a\ddot{a} - k - \dot{a}^2\right),$$
$$K^{0}_{\theta\theta 00} = \left(\dot{a}\ddot{a} - a\,\dddot{a}\right) r^2,\ K^{0}_{\phi\theta\phi\theta} = K^{0}_{\theta\phi\theta\phi} = a\dot{a}r^4 \sin^2\theta\left(k + \dot{a}^2 - a\ddot{a}\right),$$
$$K^{0}_{\theta\theta\phi\phi} = K^{0}_{\phi\phi\theta\theta} = a\dot{a}r^4 \sin^2\theta\left(a\ddot{a} - k - \dot{a}^2\right),\ K^{0}_{\phi\phi 00} = \left(\dot{a}\ddot{a} - a\,\dddot{a}\right) r^2 \sin^2\theta,$$
$$K^{r}_{00r0} = K^{\theta}_{00\theta 0} = K^{\phi}_{00\phi 0} = \frac{a\,\dddot{a} - \dot{a}\ddot{a}}{a^2},\ K^{r}_{\theta 0\theta r} = K^{r}_{\theta r0\theta} = K^{r}_{0r\theta\theta} = K^{\phi}_{\theta 0\theta\phi} = K^{\phi}_{\theta\phi 0\theta} = K^{\phi}_{0\phi\theta\theta} = \frac{\dot{a}r^2}{a}\left(a\ddot{a} - k - \dot{a}^2\right),$$
$$K^{r}_{\phi 0\phi r} = K^{r}_{\phi r0\phi} = K^{r}_{0r\phi\phi} = K^{\theta}_{\phi 0\phi\theta} = K^{\theta}_{\phi\theta 0\phi} = K^{\theta}_{0\theta\phi\phi} = K^{\theta}_{\phi\phi 0\theta} = K^{\theta}_{0\phi\theta\phi} = \frac{\dot{a}r^2}{a}\sin^2\theta\left(a\ddot{a} - k - \dot{a}^2\right),$$
$$K^{r}_{\theta 0r\theta} = K^{r}_{\theta\theta 0r} = K^{r}_{0\theta r\theta} = K^{r}_{\phi\phi 0r} = K^{\phi}_{\theta 0\phi\theta} = K^{\phi}_{0\theta\phi\theta} = K^{\phi}_{\theta\theta 0\phi} = \frac{\dot{a}r^2}{a}\left(k + \dot{a}^2 - a\ddot{a}\right),$$
$$K^{r}_{\phi 0r\phi} = K^{r}_{0\phi r\phi} = K^{\theta}_{\phi 0\theta\phi} = \frac{\dot{a}r^2}{a}\sin^2\theta\left(k + \dot{a}^2 - a\ddot{a}\right),\ K^{r}_{0r00} = K^{\theta}_{0\theta 00} = K^{\phi}_{0\phi 00} = \frac{\dot{a}\ddot{a} - a\,\dddot{a}}{a^2},$$
$$K^{r}_{\theta r\theta 0} = K^{\phi}_{\theta\phi\theta 0} = \frac{2\dot{a}r^2}{a}\left(a\ddot{a} - k - \dot{a}^2\right),\ K^{r}_{\phi r\phi 0} = K^{\theta}_{\phi\theta\phi 0} = \frac{2\dot{a}r^2}{a}\sin^2\theta\left(a\ddot{a} - k - \dot{a}^2\right),$$
$$K^{r}_{\theta\theta 00} = \frac{2\dot{a}r^2}{a}\sin^2\theta\left(k + \dot{a}^2 - a\ddot{a}\right),\ K^{r}_{\phi\phi r0} = K^{\theta}_{\phi\phi\theta 0} = \frac{2\dot{a}r^2}{a}\sin^2\theta\sin^2\theta\left(k + \dot{a}^2 - a\ddot{a}\right),$$
$$K^{\theta}_{r0\theta r} = K^{\theta}_{0r\theta r} = K^{\theta}_{rr0\theta} = K^{\phi}_{r0\phi r} = K^{\phi}_{0r\phi r} = K^{\phi}_{rr0\phi} = \frac{\dot{a}}{a\left(1-kr^2\right)}\left(k + \dot{a}^2 - a\ddot{a}\right),$$
$$K^{\theta}_{r0r\theta} = K^{\theta}_{r\theta 0r} = K^{\theta}_{0\theta rr} = K^{\phi}_{r0r\phi} = K^{\phi}_{r\phi 0r} = K^{\phi}_{0\phi rr} = \frac{\dot{a}}{a\left(1-kr^2\right)}\left(a\ddot{a} - k - \dot{a}^2\right),$$
$$K^{\theta}_{rr\theta 0} = K^{\phi}_{rr\phi 0} = K^{\phi}_{\theta\theta\phi 0} = K^{\phi}_{r\phi r0} = \frac{2\dot{a}}{a\left(1-kr^2\right)}\left(k + \dot{a}^2 - a\ddot{a}\right),\ K^{\theta}_{r\theta r0} = \frac{2\dot{a}}{a\left(1-kr^2\right)}\left(a\ddot{a} - k - \dot{a}^2\right)$$

Ricci Tensor

$$R_{\alpha\beta} = R^{\gamma}_{\alpha\gamma\beta}$$

$$R_{00} = -\frac{3\ddot{a}}{a},$$
$$R_{rr} = \frac{a\ddot{a} + 2k + 2\dot{a}^2}{1-kr^2},$$
$$R_{\theta\theta} = r^2\left(a\ddot{a} + 2k + 2\dot{a}^2\right),$$
$$R_{\phi\phi} = r^2\sin^2\theta\left(a\ddot{a} + 2k + 2\dot{a}^2\right)$$

Ricci Scalar

$$R = R^{\alpha}_{\alpha} = g^{\alpha\beta} R_{\alpha\beta}$$

$$R = \frac{6}{a^2}\left(a\ddot{a} + k + \dot{a}^2\right)$$

Cosmological Principle

$$\text{Cosmological Principle} \Rightarrow K^{\alpha}_{\beta\gamma\mu\nu} = 0$$

$$\begin{aligned}
&k + \dot{a}^2 - a\ddot{a} = 0 \wedge \dot{a}\ddot{a} - a\dddot{a} = 0 \Rightarrow \\
&a = \frac{1}{2X}\left(Ye^{Xct} + \frac{k}{Y}e^{-Xct}\right) \\
&\dot{a} = \frac{1}{2}\left(Ye^{Xct} - \frac{k}{Y}e^{-Xct}\right) = \sqrt{X^2a^2 - k} \\
&\ddot{a} = \frac{X}{2}\left(Ye^{Xct} + \frac{k}{Y}e^{-Xct}\right) = X^2 a \\
&\dddot{a} = \frac{X^2}{2}\left(Ye^{Xct} - \frac{k}{Y}e^{-Xct}\right) = X^2\sqrt{X^2a^2 - k} \\
&[X] = m^{-1},\ [Y] = m^{-1}
\end{aligned}$$

Einstein Tensor

$$G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R$$

$$\begin{aligned}
&G_{00} = \frac{3}{a^2}\left(k + \dot{a}^2\right) = 3X^2, \\
&G_{rr} = \frac{2a\ddot{a} + k + \dot{a}^2}{kr^2 - 1} = \frac{3X^2a^2}{kr^2 - 1}, \\
&G_{\theta\theta} = -r^2\left(2a\ddot{a} + k + \dot{a}^2\right) = -3X^2a^2r^2, \\
&G_{\phi\phi} = -r^2\sin^2\theta\left(2a\ddot{a} + k + \dot{a}^2\right) = -3X^2a^2r^2\sin^2\theta \\
&G_{\alpha\beta} = -3X^2 g_{\alpha\beta}
\end{aligned}$$

Pressure & Cosmological Constant

$$\begin{aligned}
G_{\alpha\beta} + \Lambda g_{\alpha\beta} &= \frac{8\pi G}{c^4}T_{\alpha\beta} \\
T_{\alpha\beta} &= \left(\rho + \frac{P}{c^2}\right)u_\alpha u_\beta + Pg_{\alpha\beta} \\
u^\alpha e_\alpha &= ce_0 \Rightarrow u_\alpha e^\alpha = -ce^0
\end{aligned}$$

$$\begin{aligned}
\left(-3X^2 + \Lambda\right)g_{\alpha\beta} &= \frac{8\pi G}{c^4}\left(\left(\rho + \frac{P}{c^2}\right)u_\alpha u_\beta + Pg_{\alpha\beta}\right) \\
\rho &= \frac{c^2}{8\pi G}\left(3X^2 - \Lambda\right) \\
P &= \frac{c^4}{8\pi G}\left(-3X^2 + \Lambda\right) = -\rho c^2 \\
\Lambda &= 3X^2 - \frac{8\pi G}{c^2}\rho
\end{aligned}$$

Measurements and Calculations

$$
\begin{aligned}
D &:= \text{Measured Distance}, [D] = m \\
V &:= \text{Measured Speed}, [V] = \frac{m}{s} \\
ct &= 0 \rightarrow a = a_0 \\
ct &= -D \rightarrow a = a\left(-D\right) \\
1 + z = 1 + \frac{V}{c} &= \frac{a_0}{a(-D)} \\
a &= a_0 \cosh Xct + \frac{\dot{a}_0}{X} \sinh Xct \\
\frac{a(-D)}{a_0} &= \frac{1}{1+\frac{V}{c}} = \cosh XD - \frac{\dot{a}_0}{a_0 X} \sinh XD \\
\mathbf{y} &= \cosh \alpha\mathbf{x} - \beta \sinh \alpha\mathbf{x} \\
\mathbf{y} &:= \frac{1}{1+\frac{V}{c}} \wedge \mathbf{x} := D \wedge \alpha := X \wedge \beta := \frac{\dot{a}_0}{a_0 X} \\
\therefore X &= \alpha \wedge k = a_0^2 \alpha^2 \left(1 - \beta^2\right)
\end{aligned}
$$

Results

$$
\begin{aligned}
\alpha &\in \left]2.894383 \times 10^{-26},\ 3.009693 \times 10^{-26}\right[\ m^{-1} \\
\beta &\in]0.298094,\ 0.306256[\\
X^2 &\in \left]8.377454 \times 10^{-52},\ 9.058253 \times 10^{-52}\right[\ m^{-2} \\
k &\in \left]7.591712 \times 10^{-52},\ 8.253337 \times 10^{-52}\right[\ m^{-2} \\
P &\in \left]-7.783152 \times 10^{-10},\ -7.555587 \times 10^{-10}\right[\ \frac{kg}{m \times s^2} \\
\Lambda &\in \left]2.351608 \times 10^{-51},\ 2.560573 \times 10^{-51}\right[\ m^{-2}
\end{aligned}
$$

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